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ABSTRACT

Large Language Models (LLMs) are revolutionizing conversational recommender systems (CRS) by effectively indexing item content, understanding complex conversational contexts, and generating relevant item titles. However, the autoregressive nature of LLMs, which outputs item titles as a long sequence of subtokens, hinders the ability to efficiently obtain and control recommendations across the entire item set. This challenge in calculating probabilities over all items limits LLMs' potential, such as (1) limiting control over recommendation popularities and (2) preventing the synergy of marrying LLMs and traditional recommender systems (RecSys).

To address this challenge, we propose the Reindex-Then-Adapt (RTA) framework. It consists of two steps: (1) `Reindex`: a lightweight network learns to condense multi-token item titles into single tokens within the LLM and distills LLM-generated recommendations as ranked lists. This bypasses the autoregressive nature of LLMs while trying to preserve their CRS abilities; (2) `Adapt`: LLMs after reindexing enable efficient adjustment of probability distributions over single-token titles, further enhanced through RecSys integration. RTA bridges the strengths of LLMs and RecSys, enabling understanding of complex queries as LLMs do, while efficiently controlling recommended item distributions as in traditional RecSys. We show the effectiveness of our RTA over base LLMs across three CRS datasets with negligible additional parameters.

CCS CONCEPTS

• **Information systems** → **Retrieval models and ranking**; • **Computing methodologies** → *Natural language processing*.

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1 INTRODUCTION

Conversational Recommender Systems [14, 25, 38] (CRS) are an emerging recommendation task aimed at suggesting relevant and personalized items via dialogues between users and systems. Recently, Large Language Models [5, 19, 25, 33, 49] (LLMs) have demonstrated proficiency in understanding user intentions within conversational texts and exhibited substantial domain-specific knowledge (e.g., in movies). Consequently, LLMs offer distinct advantages for CRS and outperform existing non-LLM baselines [19, 25, 56], positioning LLMs as an indispensable component of CRS.

In this work, we first provide an analysis of LLMs as CRS to underpin our framework. Specifically, we view LLMs as Differentiable Search Indexing (DSI) [10, 51] models, then show LLMs' **abilities** and **limitations** for *item-indexing* and *item-recommendation* tasks:

- **Abilities**: LLMs have already indexed numerous popular movies, potentially adequate to understand complex conversation contexts and address many movie conversational recommendation scenarios.
- **Limitations**: LLMs often exhibit misalignment with (potentially rapidly changing) data distributions on target platforms. Meanwhile, LLMs typically struggle to capture collaborative information specific to a given CRS platform.

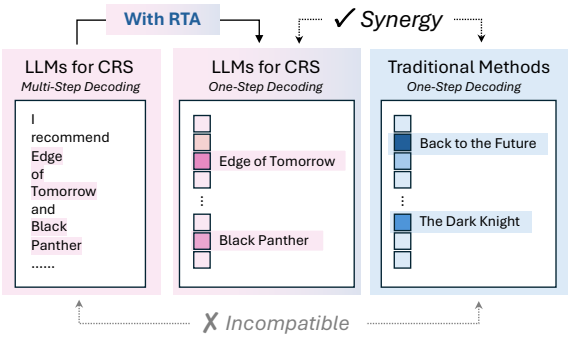


Figure 1: Illustration of Reindex-Then-Adapt (RTA) Framework. With RTA, items recommended by LLMs are reindexed as single tokens rather than multi-token titles, making calculating probabilities or logits over all items efficient. RTA eases the recommendation distribution control and enables the synergy between LLMs and traditional methods in CRS.

That is to say, while LLMs demonstrate potential for CRS tasks, they still struggle with distribution adjustments and capturing collaborative information (see Section 2), which are crucial for recommendations. Although traditional methods may offer some help in addressing these limitations, integrating them into LLMs remains challenging. We argue that this challenge stems from the *generative retrieval* paradigm of LLMs for CRS [25], i.e., generating multi-token item titles autoregressively. Specifically, for conversational recommendations, LLMs represent (i.e., index) each item with typically multiple tokens. For example, in Figure 1, the movie *Edge of Tomorrow* is usually represented by three tokens: `|Edge|`, `|of|`, and `|Tomorrow|`. LLMs generate multi-token titles sequentially and may potentially mix in textual words, instead of producing a probability or logit vector over all items in one step. While this *generative retrieval* paradigm is useful [25], it also makes obtaining recommendation probabilities for a large-scale set of items extremely inefficient, thus hindering the potential of LLMs in CRS.

To address this issue, we propose a *Reindex-Then-Adapt (RTA)* framework. We first treat LLMs as DSI models and verify our assumption that off-the-shelf LLMs have already indexed sufficient items (at least for the movie domain). Subsequently, we conduct two steps: **`reindex`** and **`adapt`**. (1) In `reindex` step, we aim to let LLMs re-index items as single-token embeddings instead of original multi-token titles, meanwhile preserving CRS capabilities. We achieve this by using lightweight networks to distill multi-token item titles (e.g., *Edge of Tomorrow*) into single tokens (e.g., `|Edge_of_Tomorrow|`) from LLM-generated recommendations. (2) In `adapt` step, we are allowed to efficiently obtain the recommended item probabilities (or logits) from the reindexed LLMs, facilitating various adaptation explorations, such as *bias terms adjustment* or *RecSys gating* (Section 3.3) to further improve CRS accuracy metrics.

Based on the *RTA* framework, we investigate three reindexing modules and two adaptation strategies across three CRS datasets. Our experimental results show improved recommendation accuracy in CRS. Additionally, our studies highlight the significance of adjusting LLMs towards target distributions and the synergistic benefits of combining LLMs and traditional RecSys in CRS.

2 PRELIMINARIES

2.1 Task Formulation

In CRS, a conversation is represented by $C = (u_t, s_t, \mathcal{I}_t)_{t=1}^{T}$ involving users $u_t \in \mathcal{U}$ and items $\mathcal{I}_t \subseteq \mathcal{I}$ with T conversation turns. Each utterance s_t comprises tokens v_i from vocabulary $\mathcal{V}$. A conversation typically involves a *seeker* and a *recommender*. As formulated in CRS studies [9, 25, 38, 57, 66], our goal is to learn a recommender to generate a ranked list of items $\hat{\mathcal{I}}_k$ at turn k that aligns with $\mathcal{I}_k$, based on the preceding context $(u_t, s_t, \mathcal{I}_t)_{t=1}^{k-1}$.

2.2 Differentiable Search Index (DSI)

The transformer model has shown proficiency in retrieval tasks, encoding item information within its parameters, which is a method termed Differentiable Search Index (DSI) [51]. DSI involves two key training tasks for pre-trained language models: *Learn to Index* (`L2I`) and *Learn to Retrieve* (`L2R`), which can be used to train a model jointly or in sequential order. `L2I` focuses on mapping item content, such as movie description, to item indices, exemplified by linking a description of *Edge of Tomorrow* to its title:

> **L2I Example**: *"A 2014 American science fiction action film starring Tom Cruise and Emily Blunt with ..."* → **Edge of Tomorrow**

`L2R`, on the other hand, maps queries to item indices, such as:

> **L2R Example**: *"I'm feeling bored today and looking for a sci-fi action movie, preferably starring Tom Cruise."* → **Edge of Tomorrow**

DSI models are originally proposed for text retrieval tasks [51], yet their formulation can be connected to LLMs used in CRS. Considering the *LLMs as CRS* framework proposed in [25] through the lens of DSI, we observe that:

- **Item Indexing**: LLMs index items by using the item titles (e.g., *"Edge of Tomorrow"*) as the item identifiers via `L2I`.
- **Item Recommendation**: LLMs use conversational context as queries to generate item indices via `L2R`.

Thus, LLMs inherently function as DSI models, by including a certain number of training samples for `L2I` and `L2R` tasks in their pre-training corpus. Compared to common two-tower models, DSI models require only a single model for item recommendations, by indexing item information into its parameters [51].

2.3 Item-Indexing Ability Analysis

According to [25], LLMs have superior knowledge in content and context, particularly in the movie domain. This proficiency is attributed to the performance in the "Learn to Index" (`L2I`) task via DSI [51]. Here, our primary question is the extent to which item content has been indexed in LLMs through the pre-training corpus.

Finding: LLMs Show Sufficient Item Content Knowledge. We gathered 6,281 pairs of movie titles from ReDIAL [38] and the related descriptions from Wikipedia[1] for experiments in Figure 2 to assess LLMs performance on `L2I` task, and observe that: First, all LLMs had indexed a considerable amount of movie content for conversational recommendation tasks. Notably, for frequently

[1] https://www.wikipedia.org/

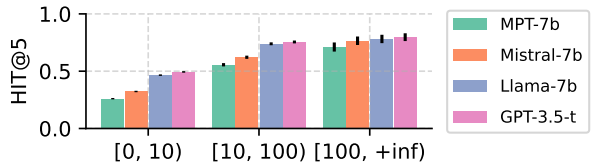


Figure 2: Item Indexing tasks by using movie descriptions from WikiPedia [3] to prompt movie titles. We tested `MPT-7b`, `Mistrial-7b` , `Llama2-7b` and `GPT-Turbo-3.5` models and group the accuracy by the range of occurrences of the movies in ReDIAL CRS dataset [38]. We measure the performance with HIT@5, i.e., whether the target movie in the top-K movie list generated by the LLMs to reflect the movie knowledge stored in the parameters of the LLMs.

Table 1: Additional statistics for cold items, warm items and popular items in ReDIAL [38] datasets. #Items counts the total numbers of items and #Occurences sums the total occurences of such items in conversations.

ReDIAL [38]	**Cold – [0, 10)**	**Warm – [10, 100)**	**Pop. – [100, +inf)**
#Items	**4,960 (78.97%)**	1,193 (18.99%)	128 (2.04%)
#Occurences	12,523 (18.03%)	**33,304 (47.94%)**	**23,647 (34.04%)**

mentioned movies, as defined as movie occurences in the ReDIAL dataset [38] that the item occurences range is [100, +inf), all LLMs exhibit impressive content knowledge. Second, the proprietary model GPT-3.5-t [49] outperforms others. Among the open-sourced LLMs of similar size, Llama2 demonstrates the best performance in the given task, as shown in Figure 2, making it the chosen base model for our subsequent experiments.

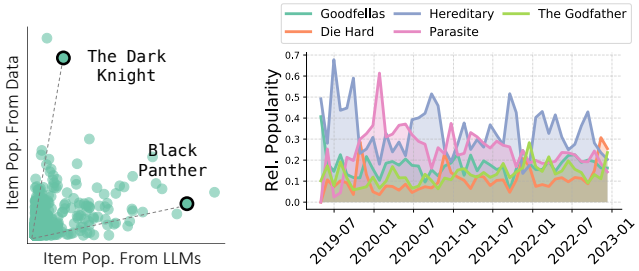
LLMs without specific fine-tuning have already indexed numerous popular movies. Meanwhile Table 1 shows the imbalance of item occurrences in conversational recommendations, where items labeled as *warm* and *pop.* constitute about 20% in terms of item counts but contribute to over 80% of recommendation occurrences. This suggests that **off-the-shelf LLMs may store sufficient item knowledge for handling many movie conversational recommendation scenarios, which opens up the opportunities of our `reindex` step**[2].

2.4 Item-Recommendation Ability Analysis

Even though LLMs can index items effectively, there remain some limitations in LLMs for CRS given a specific platform.

Finding: LLMs Show Severe Distribution Misalignment. We use item popularity as an example to show this *distribution misaligment* limitation from two perspectives. The first is a static perspective: As depicted in Figure 3(a), LLMs reflect item popularities from the large-scale training corpus to some extent, which often do not align with popular items on the specific platform for CRS. Also, from the dynamic perspective, target data distributions, such as item popularity, undergo rapid changes over time due to factors like

[2]Although, we admit fine-tuning LLMs to cover more cold items remains future work.



(a) Static Item Popularities (b) Item Popularities Over Time

Figure 3: (a) Item frequencies from CRS dataset (ReDIAL [38]) and from LLMs (Llama-7B [53]) to show static popularity misalignments, where *The Dark Knight* and *Black Panther* are two examples. (b) Visualization of item monthly relative popularity from Reddit-Movie [25] datasets, since this dataset is the only CRS dataset with long-range timestamps in the wild. Item popularities are shown to change over time rapidly.

seasons and promotion strategies. For instance, Figure 3(b) shows that monthly relative item popularities on the *Reddit-Movie* [25] dataset change over time, which cannot be captured by a static LLM, even fine-tuned ones.

Finding: LLMs Struggle to Use Collaborative Information. According to findings in [25], masking or randomly replacing mentioned movies in conversation contexts for CRS datasets like INSPIRED [24], ReDIAL [38], and Reddit [25] has little negative impact on the final recommendation accuracy produced by LLMs. This experiment indicates that LLMs primarily leverage textual information in conversations (e.g., movie content or user context) for recommendations and underutilize collaborative information.

Those findings suggest that, despite LLMs exhibiting impressive performance in CRS [25], there exists **room for improving recommendation accuracy by adapting towards the distributions of target platforms**; further, due to the dynamic nature of recommendation platforms, target data distributions change rapidly, necessitating the **more efficient methods to obtain and control item recommendations from LLMs accordingly**.

3 FRAMEWORK

3.1 Overview

As discussed in Section 1, we argue that representing target items with varying token counts in LLMs poses challenges to obtaining probabilities over all items and adapting LLMs to target data distributions. To tackle this, we view LLMs as DSI models that have already indexed sufficient item content knowledge (see Section 2.3), and propose the *Reindex-Then-Adapt (RTA)* framework in Figure 4:

(1) **Reindex** item indices with varying token numbers in LLMs into single-token item indices using a mixture of data samples from the `L2I` corpus and `L2R` corpus, generated by LLMs. This aims to remove the `adapt` step barrier. In contrast to the `index` step in the original DSI models [51], the **`re`**`index` step reuses the content of indexed items from the LLMs, thereby

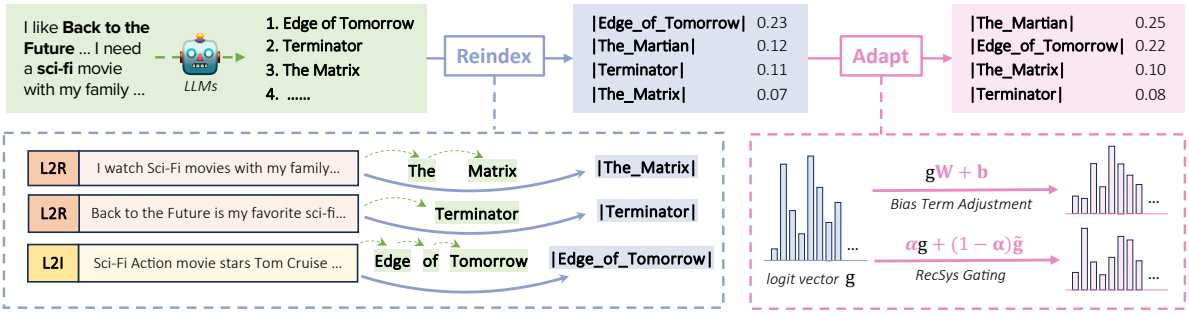


Figure 4: Reindex-Then-Adapt (RTA) Framework. LLMs can generate a list of item titles as the recommendations given the conversation contexts. To further improve the accuracy and controllability, we conduct (1) `reindex` step: reindexing the item (e.g., movie) titles in LLMs as single tokens to obtain the predicted *logit* vectors efficiently; (2) `adapt` step: adapting the recommenders towards target data distributions effectively with multiple options on the *logit* vectors such as adjusting bias terms or combining RecSys models with Gating mechanism.

facilitating the learning process for the new item indices, so the reindexing network can be lightweight.

(2) **Adapt** *logits* from the reindexed LLMs, achieved by transforming the *logit* vectors or combining with other traditional RecSys using Gating mechanism [15, 22, 28]. This adjustment aims to effectively adapt the recommendation probability distributions over items towards the target data distributions.

3.2 `Reindex` Step: Single-Token Items in LLMs

The key of `reindex` step is to condense multi-token item embeddings into single-token item embeddings efficiently, and preserve the LLMs' abilities for CRS tasks.

3.2.1 Identify Item Indices From LLMs-Generated Responses. First, we prompt LLMs to generate responses with mentioned items at scale, to construct a corpus $\mathcal{S}$, where each sentence $s \in \mathcal{S}$ is the question from the seed corpus and answer from the LLMs. More details about the seed corpus will be described in Section 3.2.4. Formally, for a sentence $s = (v_i)_{i=1}^{m}$ consisting of tokens $v_i \in \mathcal{V}$, we denote the tokens representing an item in the LLM responses as $(v_i)_{i=j}^{j+n}$, where j indicates the starting position of the first token for the item in s, and n is the number of tokens representing this item. Consequently, with the `Embed` layer, LLMs can retrieve the sequence of token embeddings for this item:

$$(\mathbf{v}_i)_{i=j}^{j+n} = \texttt{Embed}\left((v_i)_{i=j}^{j+n}\right), \tag{1}$$

where $\mathbf{v} \in \mathbb{R}^d$ is the token embedding, and d is the embedding size. For example, we may look up *"Edge of Tomorrow"* embeddings represented by "[14, 72, 98]" in the sentence s.

3.2.2 Aggregate Multi-Token Embeddings. In the proposed `reindex` step, we assume that the semantics from multiple (typically shorter than 10) token embeddings can be aggregated into a new single token embedding with a trainable aggregator, such as:

$$\tilde{\mathbf{v}} = \texttt{Aggregator}\left((\mathbf{v}_i)_{i=j}^{j+n}\right), \tag{2}$$

where this aggregator is treated as our reindexing module, and the aggregated $\tilde{\mathbf{v}} \in \mathbb{R}^d$ serves as the new representation of the target item in LLMs generation. Therefore, if all target items are represented by the "squeezed" single embedding, scoring all the target items to obtain a logit vector from LLMs is efficient. In general, many existing model architectures can be used as `Aggregator`, such as RNN-based [11], Transformer-based models [54], or even Weighted Pooling (see more details in Section 4.3).

3.2.3 Learning Process. The contrastive loss [46] is used for learning the aggregator to "condense" multi-token item embeddings and preserve the semantics for LLMs in CRS tasks:

$$\mathcal{L}_{reindex} = -\frac{1}{|\mathcal{D}|} \sum_{\mathbf{q},\tilde{\mathbf{v}} \in \mathcal{D}} \log \left[\frac{\exp\left(\mathbf{q}^\top \tilde{\mathbf{v}}\right)}{\exp\left(\mathbf{q}^\top \tilde{\mathbf{v}}\right) + \sum_{\mathbf{n} \in \mathcal{N}} \exp\left(\mathbf{q}^\top \mathbf{n}\right)} \right], \tag{3}$$

where we loop over the training set $\mathcal{D}$ that consists of $(\mathbf{q}, \tilde{\mathbf{v}})$ pairs extracted from LLM-generated data $\mathcal{S}$. Here, $\mathbf{q} \in \mathbb{R}^d$ is the contextual embedding of the last position from a LLM, which is originally used to generate the first token of original indexed items, but now we aim to force the aggregated item representation $\tilde{\mathbf{v}} \in \mathbb{R}^d$ described in Section 3.2.2 to be generated by LLMs. To achieve this reindex step, we also prepare random negatives $\mathbf{n} \in \mathbb{R}^d$ from the negative representation set $\mathcal{N}$ of other reindexed item titles.

3.2.4 Data Sources. We consider mixing two groups of data sources as a unified corpus and using it to train our model jointly.

- **L2R Data**: The (query, target) pairs are treated as `L2R` samples. In the CRS context, those queries are from the conversations, and targets are from LLM-generated responses.
- **L2I Data**: The (content, target) pairs are treated as `L2I` samples. In the CRS context, those contexts are samples from the item metadata like textual descriptions, and targets are identified from LLM-generated responses.

Note that our `reindex` method can be treated as a specially designed *distillation* method [27], but the major benefit is that an additional (perhaps large) student model is not required. Instead, we only introduce a lightweight network. For example, the RNN reindexing network used in our experiments has only 0.43% of the original LLMs parameter counts. More distillation methods and data synthesis strategies can be explored further.

3.3 Adapt Step: Item Probabilities Adjustment

After re-indexing, all the items are represented by single-token embeddings. It makes a recommendation as easy as one-step decoding in LLMs and also enables multiple efficient ways to adjust the recommendation item distributions to adapt towards target platforms or specific data distributions or combine them with additional models efficiently. We introduce two types of adaptation methods in the following sections, one for item popularity adjustments, and another one for combining with the traditional recommender systems. To start with, we assume the logit vector $\mathbf{g} \in \mathbb{R}^{|\mathcal{I}|}$ has already been given by the LLMs, and the corresponding probability vector should be $\mathbf{p} = \texttt{softmax}(\mathbf{g})$.

3.3.1 Bias Term Adjustment. Inspired by [64], a common way to adjust logits is an affine transformation, i.e.:

$$\hat{\mathbf{p}} = \texttt{softmax}\left(\mathbf{g}\mathbf{W} + \mathbf{b}\right), \tag{4}$$

where $\mathbf{W} \in \mathbb{R}^{|\mathcal{I}|\times|\mathcal{I}|}$ is a weight matrix and $\mathbf{b} \in \mathbb{R}^{|\mathcal{I}|}$ is the bias term. Similar to [64], we restrict the matrix $\mathbf{W}$ to be diagonal to prevent the size of parameters from growing quadratically in the size of items. Therefore, in this special case, we are able to interpret the $\mathbf{W}$ and $\mathbf{b}$ as *multiplicative* and *additive bias* terms towards the target data distributions, respectively.

3.3.2 Traditional RecSys Gating. Inspired by [25], we notice traditional RecSys, where the output logit vector can be denoted as $\tilde{\mathbf{g}} \in \mathbb{R}^{|\mathcal{I}|}$, is good at collaborative knowledge instead. Motivated by this observation, combining LLMs and traditional RecSys becomes easy after the reindex step:

$$\hat{\mathbf{p}} = \texttt{softmax}\left(\alpha\mathbf{g} + (1-\alpha)\tilde{\mathbf{g}}\right), \tag{5}$$

where the coefficient $\alpha \in [0, 1]$ can be set in many different ways. For simplicity sake, we use a learnable scalar $\tilde{\alpha}$ for $\alpha = \texttt{sigmoid}\left(\tilde{\alpha}\right)$ in our experiments. Note that more options can be considered, such as being predicted by an MLP model to naturally determine how much to weight the responses from LLMs or traditional RecSys.

3.3.3 Learning Process. We use maximum likelihood estimation to derive the loss for `adapt` step, in order to learn the parameters of the bias terms or the recsys model. Note that the LLMs parameters are not involved in this step, ensuring an efficient learning process:

$$\mathcal{L}_{adapt} = -\frac{1}{|\mathcal{D}^*|}\sum_{i=1}^{|\mathcal{D}^*|}\log \hat{\mathbf{p}}_{i,*}, \tag{6}$$

where dataset $\mathcal{D}^*$ is collected from the target platform, such as ReDIAL [38], which is typically a small sized dataset. Here, $\hat{\mathbf{p}}_{i,*}$ denotes the probability of the ground-truth item in the ith data sample. Our purpose is to adapt the model towards the underlying data distributions of $\mathcal{D}^*$ through this learning process.

4 EXPERIMENTS

4.1 Experiment Setup

4.1.1 Datasets. Three conversational recommendation datasets [24, 25, 38] are used in our experiments, where the statistics are summarized in Table 2: **INSPIRED** [24] and **ReDIAL** [38]: These two datasets consist of small-scale human-human conversations for movie recommendations with crowd-sourced annotations from MTurk. In the following experiments, we randomly split the datasets into training, validation, and test sets using an 8:1:1 ratio. **Reddit-V1.5** [25]: This dataset comprises large-scale movie discussions on Reddit, which were collected and processed by [25]. This dataset shows movie conversation recommendations in the wild and includes corresponding timestamps for 10 years to study temporal patterns. For data splitting, we use the last two months (i.e., Nov. and Dec. in 2022) as validation and testing set respectively to approximate the real setting. Due to the large size of the given dataset, we uniformly sample 20% conversation turns for validation (i.e., 11,241 samples) and testing (i.e., 13,816 samples).

Table 2: Dataset Statistics. We update *Reddit-Movie* CRS dataset as Reddit-V1.5 according to the raw data dump provided by [25] from 2012 to 2022. The conversation turns with valid recommended items are denoted as *R_Turns*.

	INSPIRED [24]	**ReDIAL** [38]	**Reddit-V1.5** [25]
#Conv.	999	11,348	2,726,471
#Turns	21,124	139,557	5,063,007
#R_Turns	1,950	30,322	1,787,050
#Users	999	764	520,913
#Items	1,472	6,281	68,285

4.1.2 Baselines. We consider four groups of baseline models for comparison. (1) We consider some representative traditional *item-based*[3] RecSys models, **Popularity**, **FISM** [31] and **SASRec** [32]. (2) We consider some representative CRS models: **ReDIAL** [38] and **UniCRS** [57]. (3) We consider some zero-shot open-sourced LLMs as baselines like [25] and use the 7-billion-parameter version due to compute burden: **MPT-7b** [52], **Mistral-7b** [30] and **Llama2-7b** [53]. Those zero-shot LLMs are considered powerful CRS models in [25]. (4) We consider some dense retrieval models given the connections to document retrieval: **SBERT** [47] and **Instructor** [50]. We also discuss the results from **GPT-3.5-turbo** [49], which is a much larger proprietary model that can achieve state-of-the-art CRS performance even in a zero-shot setting [25].

4.1.3 Evaluation Metrics. We focus on recommendation accuracy HIT@K (H@K) and NDCG@K (N@K), following [9, 38, 57, 66]. We consider the means and the standard errors[4] of the metrics with $K = \{5, 10\}$[5].

4.2 General CRS Performance

4.2.1 Baseline Performance. Table 3 shows the recommendation accuracy of four groups of baselines on three conversational recommendation datasets. There are some observations:

On Traditional RecSys. Conventional recsys models effectively capture target popularity and further item-item similarities. Interestingly, on INSPIRED, we find that non-personalized popularity serves as a strong baseline, because the limited size of the training set may restrict the ability to capture more complex item-item relationships. The results of traditional recommendation system

[3]We only use item-based models since INSPIRED does not have user histories.
[4]We use error bars in our figures and gray numbers in our tables for standard errors.
[5]More details of our setup and experiment results can be found in https://github.com/AaronHeee/Reindex-Then-Adapt.

Table 3: The main results for our models on conversational recommendation accuracy performance, compared against (1) traditional recommendation models; (2) zero-shot large language models (LLMs); (3) traditional conversational recommendation models; and (4) zero-shot dense retrievers. The size of the reported LLMs used here is 7B. We denote the model metrics with the best performance in bold. Llama2-R denotes the Llama2-7b model after our `reindex` step. We also show the results after the `adapt` step with bias terms (+Bias) or RecSys (here we use FISM [31]) model combination with Gating mechanism (+RecSys).

	INSPIRED				ReDIAL				Reddit-V1.5			
	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10
Popuplarity	.089 .020	.065 .015	.103 .021	.070 .015	.035 .003	.025 .002	.052 .003	.030 .002	.008 .001	.004 .000	.014 .001	.006 .000
FISM [31]	.090 .017	.065 .012	.104 .021	.072 .011	.065 .004	.040 .003	.112 .005	.054 .003	.022 .001	.012 .001	.043 .001	.019 .001
SASRec [32]	.092 .014	.068 .011	.103 .020	.073 .013	.068 .004	.041 .002	.116 .005	.056 .003	.022 .001	.013 .001	.039 .001	.018 .001
MPT [52]	.075 .018	.045 .011	.099 .020	.052 .012	.072 .004	.045 .003	.116 .005	.059 .003	.026 .001	.017 .001	.040 .001	.021 .001
Mistral [30]	.061 .016	.040 .011	.066 .017	.041 .012	.082 .004	.056 .003	.111 .005	.065 .003	.029 .001	.020 .001	.038 .001	.023 .001
Llama2 [53]	.080 .019	.050 .012	.122 .022	.064 .013	.094 .004	.059 .003	.145 .005	.075 .003	.042 .001	.027 .001	.064 .001	.034 .001
ReDIAL [38]	.060 .016	.041 .012	.106 .021	.056 .012	.067 .004	.044 .003	.106 .005	.057 .003	.029 .001	.019 .001	.044 .001	.024 .001
UniCRS [57]	.091 .019	.055 .011	.132 .019	.073 .014	.085 .003	.058 .003	.112 .004	.071 .003	.028 .001	.017 .001	.040 .001	.021 .001
SBERT [47]	.038 .013	.026 .010	.066 .017	.036 .010	.016 .002	.010 .001	.026 .002	.013 .001	.003 .000	.002 .000	.005 .000	.002 .000
Instructor [50]	.052 .015	.034 .011	.085 .019	.045 .011	.025 .002	.013 .001	.043 .003	.019 .001	.009 .001	.006 .000	.017 .001	.008 .000
Llama2-R	.075 .018	.045 .012	.131 .023	.063 .012	.072 .004	.044 .003	.120 .005	.059 .003	.034 .001	.020 .001	.056 .001	.028 .001
w/ Bias	.117 .022	.077 .016	.174 .026	.095 .016	.096 .004	.061 .003	**.165 .006**	.083 .003	.052 .001	.033 .001	.088 .002	.044 .001
w/ RecSys	**.131 .023**	**.081 .015**	**.197 .027**	**.101 .016**	**.103 .005**	**.066 .003**	**.165 .006**	**.085 .003**	**.057 .001**	**.035 .001**	**.093 .002**	**.047 .001**

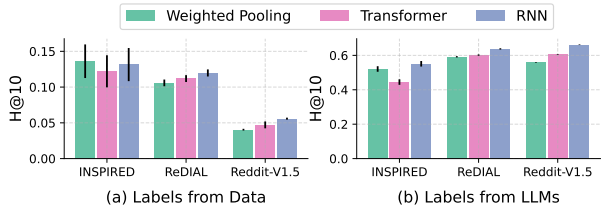


Figure 5: Different methods to represent items in LLMs with single-token embeddings and the related recommendation accuracy HIT@10 after the `reindex` step.

models also indicate the potential of improving the recommendation accuracy by aligning with target data distributions.

On LLMs. LLMs with zero-shot prompting from [25] achieve impressive results, surpassing even the best baselines on ReDIAL datasets. Additionally, the rank of recommendation accuracy within the LLM group aligns with the performance from Figure 2.

On Other Baselines. We observe that zero-shot state-of-the-art dense retrievers are unable to achieve comparable performance as zero-shot LLMs; this may be due to two reasons: (1) Dense retrievers focus more on retrieving similar documents according to semantic similarities (e.g., similar contents), but LLMs show better understanding abilities for conversation contexts; (2) We are encoding the movie textual title rather than the description of the movie for fair comparison, which may limit the dense retrievers' performance. As for traditional CRS models, since we follow the setting in [25] to remove "repeated" items, many popular CRS models perform relatively weaker in the corrected evaluation protocol.

4.2.2 Ours vs. Baselines. We construct a small-sized aggregator on top of Llama2 as an example, then use this aggregator to reindex multi-token movie titles into single-token item titles as recommendation candidates, namely Llama2-R.

On Recommendation Accuracy. Table 3 shows the RTA framework excels over baselines on those three datasets. From the `reindex` step (Llama2-R) and `adapt` step (+Bias or +RecSys), we observe a potential performance decrease in the `reindex` step due to the distillation performance gap from the original token embeddings to the single token embeddings from the relatively small aggregator. However, our models compensate by capturing the target data distribution through bias terms or traditional RecSys models.

On Efficiency and Flexibility. In terms of parameter counts, the aggregator-based methods are around 12.8% compared to an out-of-vocabulary item embedding tables and 0.43% compared to the Llama2-7b base model, emphasizing its space efficiency. Additionally, as all movie titles with varying token length are "squeezed" into single tokens, our model can rank all items with a single decoding step, making it around 100× faster than the generative retrieval from LLMs to recommend the top-20 items in our experiments.

4.3 Effectiveness of the `Reindex` Step

4.3.1 Experiment Setup. We explore three aggregator architectures for reindexing: (1) **Weighted Pooling**: learning position-wise attention weights to aggregate multi-token embeddings into a single one, followed by a simple linear projection; (2) **Transformer** (TRM): employing a single-layer transformer to derive a contextual embedding from the output `CLS` token; (3) **RNN**: using a simple GRU model to aggregate multiple token embeddings, with the last hidden state vectors serving as the item representations. Meanwhile, we calculate the hit ratios (H@10) between aggregator outputs and two label sets in evaluation: (1) **Labels From Data**: to what extent we hit targeted items from human conversations in CRS datasets; (2) **Labels From LLMs**: to what extent we hit targeted items generated by Llama2.

4.3.2 Different Aggregators. Among the three aggregators in Figure 5, the Weighted method demonstrates competitive performance

Table 4: Recommendation accuracy comparison among Llama2-R, and the detailed configurations of adding bias terms or RecSys gating.

	Agg. Frozen?	INSPIRED	ReDIAL	RedditV1.5
Llama2-R	–	.131 .023	.120 .005	.056 .001
Bias Term Adjustment (w/ Bias)				
w/ gW	×	.150 .025	.165 .006	.082 .002
	✓	.146 .024	.141 .006	.058 .001
w/ b	×	.174 .026	.165 .006	.087 .002
	✓	.136 .235	.117 .005	.058 .001
w/ gW+b	×	.155 .025	.167 .006	.088 .002
	✓	.160 .025	.144 .005	.058 .001
RecSys Model Gating (w/ RecSys)				
w/ FISM	×	.197 .027	.146 .005	.093 .002
	✓	.207 .028	.165 .006	.074 .002
w/ SASRec	×	.178 .026	.148 .005	.093 .002
	✓	.188 .027	.157 .006	.075 .002

despite its simple architecture. This suggests that the existing token embeddings from the LLMs are effective enough, making the weighted sum with linear projection a reasonable approach to consolidating token embeddings. Additionally, TRM performs worse than RNN, possibly because (1) titles (e.g., movies) are typically short (fewer than 20 tokens), diminishing the significance of TRM's advantages over RNN in handling long dependencies; (2) `CLS` tokens show difficulty in representing a sentence, as noted in [12].

4.3.3 Different Evaluation Sets. Figure 5 shows that those three aggregators all show high agreement (around 62% H@10) with items generated by LLMs, which suggests that our aggregator can distill the LLMs' behaviors of making recommendations to a large extent. Meanwhile, we observe that, within datasets, the aggregator with a higher agreement with LLM-generated recommendations tends to achieve higher recommendation accuracy on this dataset, which also indicates the effectiveness of our reindex step.

4.4 Effectiveness of the `Adapt` Step

4.4.1 Component Analysis. Table 4 shows introducing bias terms after the `reindex` step is a simple yet effective strategy. This is attributed to the potential for improving recommendation accuracy by addressing popularity misalignments, as discussed in Section 2.4. Meanwhile, Table 4 demonstrates that introducing traditional RecSys models is effective to adapt the recommendation distribution. For example, on INSPIRED, the best method with bias adjustment is 0.174 H@10, but the recsys gating method further improves the accuracy to 0.207 H@10.

4.4.2 Impact of Bias Term Types. Table 4 shows both multiplicative and additive biases improve accuracy, though their impact varies. We recommend considering both bias types empirically.

4.4.3 Impact of RecSys Model Types. Our current focus is on "item-based" RecSys models without incorporating long-term user representations. In this context, FISM and SASRec exhibit enhanced

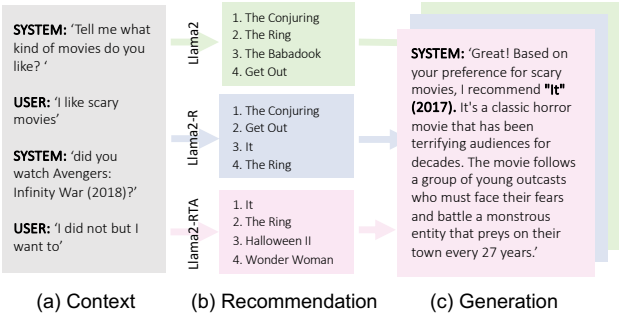


Figure 6: An Example with real results from Llama2, Llama-R and Llama-RTA (w/ FISM), followed by a natural language response from Llama2 (detailed prompts can be found in ??). This conversational context is from ReDIAL dataset, with the ground-truth movie *IT*.

performance in Table 4. Since CRS datasets do not have strong sequential patterns in the mentioned items, a simple FISM model can achieve reasonably good improvements.

4.4.4 Impact of Frozen Aggregators. Our experimental results on the smaller datasets INSPIRED and ReDIAL show that continuous training of aggregators during the adaptation step does not yield significant performance gains. This is perhaps because the synergy between LLMs and RecSys models is sufficient to capture both content and collaborative information without the need for aggregator updates. However, on the largest dataset, RedditV1.5, continuous training of the aggregator can further improve performance by capturing complex signals from texts beyond those mentioned items.

4.5 Discussions

4.5.1 Conversational Recommendation Responses. Figure 6 illustrates the complete pipeline of generating results for conversational recommendation tasks. Our discussions are below:

On Recommendation. The outputs of the *recommendation* phase are items. In Figure 6, the three models (Llama2 and its variants under our framework) understand contexts, yielding high-quality recommendations for scary movies. Specifically, Llama2-RTA builds a connection between the superhero movie *Avengers: Infinity War* in the context and the candidate *Wonder Woman*, using item-to-item relationships modeled by the FISM [31] model. Meanwhile, we posit that while multiple recommended items align with conversation contexts, the failure to adjust for the popularity of items on the target platform (e.g., movie *IT* being popular on ReDIAL) leads to zero-shot LLMs failing to meet user interests.

On Generation. In Figure 6, the generation phase is accomplished by prompting the Llama2 model. Note that our focus in this work is solely on the technical aspects of the recommendation phase. Still, we make some observations: (1) In many cases, presenting the recommendation phase suffices for users. However, our RTA framework, which introduces only a few additional parameters without changing the weights of the original LLMs, enables the reuse of LLMs for further generating natural-language responses as shown in Figure 6; (2) In CRS, there is an ongoing debate about whether to

Table 5: Recommendation accuracy comparison. After our lightweight RTA methods, the 7B open-sourced LLM (Llama2) can match and even outperform the competitive proprietary large model ChatGPT (GPT-3.5-t) in CRS [25].

	INSPIRED		ReDIAL		Reddit-V1.5	
Model	H@10	N@10	H@10	N@10	H@10	N@10
Llama2	.122 .022	.064 .013	.145 .005	.075 .003	.064 .001	.034 .001
→ RTA w/ Bias	.174 .026	.095 .016	.165 .006	.083 .003	.088 .002	.044 .001
→ RTA w/ RecSys	.197 .027	.101 .016	.165 .006	.085 .003	.093 .002	.047 .001
GPT-3.5-t [49]	.150 .024	.089 .016	.163 .006	.089 .003	.104 .002	.055 .001

perform the recommendation or generation phase first [38, 57, 66]. Our example suggests recommendation phase comes first if the recommendation phase is frequently adjusted (a common scenario due to distribution shift). Reversing the order may lead to text-item inconsistency issues (e.g., the generated response is specifically tailored for recommended movie *IT*, leading to a mismatch with the recommendation from Llama2).

4.5.2 Comparison with Proprietary Models. To deepen our understanding of the models, we adopt the zero-shot setting in [25] to query the proprietary model GPT-3.5-t [49]. As shown in Table 5, GPT-3.5-t is a competitive model for conversational recommendations. However, a small Llama2-7B model after our RTA can match or even outperform the recommendation accuracy of proprietary GPT-3.5-t. It shows the effectiveness of our framework, and it is reasonable to guess that, given our LLM-architecture-agnostic approach, building RTA upon GPT-3.5-t (if the weights are accessible) may improve the recommendation accuracy further. Thus, our reasonable next step is working on models similar to GPT-3.5-t, such as Llama3-70b, if the required compute resources are available.

5 RELATED WORK

5.1 Conversational Recommendation (CRS)

The objective of conversational recommender systems (CRS) is to elicit user preferences and deliver tailored recommendations through interactive dialogues. Historically, CRS implementations have ranged from template-driven systems [14, 26, 35, 36, 63] to critique-based approaches [8, 39, 60]. With the evolution of natural language processing, "deep" CRS models [9, 38, 57] have been developed, enabling more natural-language interactions. Research indicates the utility of CRS models is enhanced by incorporating diverse supplementary data, such as knowledge-enriched models [9, 66] utilizing external knowledge bases [3, 42], review-centric models [44], and session/sequence-oriented models [40, 67]. UniCRS [57] uses knowledge bases [3], built on DialoGPT [62] and employing prompt tuning [5], represents a state-of-the-art CRS model on datasets like ReDIAL [38] and INSPIRED [24]. Recently, an emerging topic is to leverage LLMs in CRS, with [20, 25] introducing a novel CRS pipeline, even in the zero-shot setting [25], and [56] focusing on advanced user simulation for LLM evaluation. Our research is the first to study the distribution misalignments and the synergy between LLMs and RecSys for CRS and solutions by reindexing the multi-token item titles.

5.2 Large Language Models (LLMs)

LLMs possess a remarkable capacity for generalizing to unfamiliar tasks and areas [6, 13, 59] in zero-shot or few-shot settings. Studies have shown that scaling up LLMs can significantly enhance their performance and efficiency in downstream applications [34]. In line with these developments, LLMs have been successfully applied to various downstream tasks such as question answering, numerical reasoning, code generation, and commonsense reasoning, often without requiring gradient updates [6, 34, 41, 65]. The recommendation field has recently begun integrating LLMs, either by adapting LLM architectures [16, 21] or repurposing existing LLMs for recommendation purposes [37, 43, 55]. Our study aligns with the research line of utilizing LLMs for conversational recommendations. We specially develop a distillation method to "reindex" items in LLMs to enable the synergy between LLMs and traditional methods and alleviate the concerns of LLMs for critical recommendation aspects.

5.3 LLMs for Recommendation

There is growing interest in the academic community to harness LLMs for recommendation-related tasks. One research direction explores LLMs within conventional recommendation setup, which typically incorporate user feedback and item metadata [4, 17, 23, 29, 33, 48, 61]. This includes tasks such as rating prediction [33] and sequential recommendation [23, 29, 61]. In such contexts, employing LLMs as recommenders has shown potential, particularly in scenarios with extreme data sparsity [4] or during the cold-start phase [48]. However, they often struggle to surpass simpler baseline methods, like non-personalized popularity-based models, in standard recommendation scenarios [29, 33]. Nevertheless, enhancing existing recommender systems with features generated by LLMs has yielded improved performance [2]. Another significant research direction focuses on language-centric recommendation tasks [1, 19, 20, 25, 45]. These tasks include generating explanations for recommendations, narrative-based recommendations [45], and conversational recommendations [19, 20, 25]. LLMs exhibit proficient performance in understanding intricate textual inputs, allowing for personalized recommendation outputs. Recent investigations in conversational recommendation demonstrate encouraging outcomes leveraging LLMs, even in zero-shot configurations. Our study uses existing LLMs with minimal additional parameters, implementing the *Reindex-Then-Adapt* framework, to enable the synergy between LLMs and traditional methods in CRS, which is CRS-tailored, efficient, and can have broader benefits like fairness.

6 CONCLUSION

This study proposes a solution to mitigate the limitations of large language models (LLMs) in CRS tasks, such as distribution misalignment and failure to capture collaborative information. We conceptualize LLMs as Differential Search Index (DSI) models and introduce the *Reindex-Then-Adapt (RTA)* framework. The framework condenses multi-token item titles into single tokens within LLMs (`reindex` step) and subsequently adjusts their probability distributions or enables the synergy between LLMs and traditional RecSys (`adapt` step). Empirically, the RTA framework improves recommendation accuracy metrics across various conversational recommendation datasets and adaptation settings.

7 ETHICAL CONCERNS

We note that like many other recommender systems, our proposed RTA framework optimizes for fitting user behavioral data and thus requires further alignment to take all stake-holders of a recommender system into account [18]. For example, our model can reflect popularity bias for items in the training data, leading to unfair under-exposure for less popular items. To this end, we encourage recommender system practitioners to combine the RTA framework with existing de-biasing and fairness-promoting methods [7, 58] before system deployment.

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